PAVAN DASAM

AI Engineer | Generative AI | Agentic AI | RAG

Over 3 Years+ • Financial Insurance | Health Insurance

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LangChain & LangGraph Certified

PROFESSIONAL SUMMARY

AI Engineer with over 3 years of experience designing, building and deploying enterprise Generative AI and Agentic AI solutions across Financial Insurance and Health Insurance domains. Hands-on experience developing production-oriented RAG applications, LLM-powered assistants and multi-step agentic workflows using Python, LangChain and LangGraph. Skilled across the full GenAI application lifecycle — prompt engineering, embeddings, vector search, retrieval pipelines, tool integration, structured outputs, evaluation and guardrails — with practical exposure to FastAPI-based API development and Docker/Azure deployment. Experienced collaborating with architects, product owners, SMEs and compliance stakeholders to translate regulated, enterprise business requirements into reliable, grounded and auditable AI systems.

TECHNICAL SKILLS

Programming	Python, SQL
Generative AI	Generative AI, Large Language Models (LLMs), Prompt Engineering, RAG, Embeddings, Semantic Search, Context Management, Structured Outputs
Agentic AI	AI Agents, Tool Calling, Agentic Workflows, State Management, Human-in-the-Loop, Multi-Agent Concepts
Frameworks	LangChain, LangGraph, FastAPI, Pydantic
LLMs	OpenAI GPT Models, Azure OpenAI, Anthropic Claude
Vector Databases / Search	FAISS, ChromaDB, Azure AI Search
RAG Pipeline	Document Loading, Chunking (RecursiveCharacterTextSplitter), Embeddings, Vector Search, Top-K Retrieval, Context Augmentation, Reranking Concepts, Metadata Filtering
AI Evaluation	RAGAS, LLM-as-a-Judge, Groundedness, Faithfulness, Relevance, Hallucination Evaluation
Responsible AI / Guardrails	Prompt Injection Protection, Content Filtering, PII Handling, Output Validation, Human Escalation
APIs / Backend	REST APIs, FastAPI, JSON, API Integration
Cloud / Deployment	Azure, Azure OpenAI, Docker, Git, GitHub, CI/CD Fundamentals

PROFESSIONAL EXPERIENCE

Accenture | Agentic AI Engineer

Client: Voya Financial – Financial Insurance • Generative AI & Agentic AI • Jun 2024 – Present

PRODUCTS / PROJECTS: Autonomous Claims Processing Agent — Agentic AI ★

Context: Voya's claims operations span supplemental health, life and disability, and in 2026 brought Leave/PFML/Short-Term Disability claims administration in-house for new business. Designed a multi-agent workflow, rather than a single LLM call, to process incoming claims end-to-end.

Multi-Agent Workflow: New Claim → Intake Agent → Document Verification Agent → Policy Retrieval Agent → Eligibility Agent → Claims Analysis Agent → Fraud/Risk Agent → Recommendation Agent → Human Claims Examiner

Technologies: LangGraph, Azure OpenAI, Python, FastAPI, Azure Service Bus / Event Grid, Azure AI Search, PostgreSQL, Redis, Docker, AKS

- Designed and built a multi-agent LangGraph workflow spanning Intake, Document Verification, Policy Retrieval, Eligibility, Claims Analysis, Fraud/Risk and Recommendation agents to process disability, life and supplemental health claims.
- Developed the Intake Agent to classify incoming claims and extract claimant and policy information, and the Document Verification Agent to check received documents against claim-type requirements and flag missing items.
- Implemented the Policy Retrieval Agent using RAG over certificate/policy language to surface the applicable coverage sections for each claim.
- Developed the Eligibility Agent to validate coverage dates, plan eligibility and applicable rules via internal APIs and rules-engine integration.
- Built the Fraud/Risk Agent to flag unusual claim patterns for review, and the Recommendation Agent to assemble supporting evidence, cite policy sections, and produce a confidence-scored recommended next action for the human examiner.
- Implemented human-in-the-loop routing so claims examiners review and finalize every agent-recommended action before it is applied — e.g., a sample output flags 4/5 documents received, cites Policy Section 7.2, and recommends requesting the missing employer statement at 94% confidence, routed for human review.
- Selected LangGraph over a simple LangChain chain because the claims process required persistent state, conditional branching between agents, and human-in-the-loop checkpoints.

Voya Claims Intelligence Copilot — GenAI

Context: Voya's Claims Center supports claim initiation, document upload and status tracking across life, disability and supplemental health products. Examiners needed a faster way to understand large claim files spanning medical documents, claim forms, employer information, policy documents and correspondence.

RAG Architecture: Documents → OCR/Document Extraction → Chunking → Embeddings → Vector DB → RAG → LLM → Structured Claims Summary

Technologies: Azure OpenAI / GPT, LangChain, Python, FastAPI, Azure AI Search, Blob Storage, PostgreSQL, React, Application Insights

- Built a GenAI copilot that ingests claim documents and produces a structured summary for examiners — claim type, reason, missing documents, relevant policy sections, potential issues and a recommended next step — instead of requiring manual review of the full file.
- Implemented OCR-based document extraction, chunking and embedding generation across heterogeneous claim file formats.
- Developed RAG retrieval over policy and claims documentation to ground each generated summary in the applicable plan language.

- Built FastAPI endpoints and a React-based interface for examiners to review AI-generated summaries alongside source documents, with Application Insights monitoring.
- Applied guardrails ensuring the copilot assists examiners with information synthesis only, and does not autonomously approve or deny high-impact insurance claims.

TAO Digitals | Software Engineer

Client: Anthem – Health Insurance • Java Development & Generative AI • Mar 2023 – May 2024

GENAI ENGINEER — MemberAssist AI — GenAI Member Service Copilot

Context: Members contact support with questions such as whether a procedure is covered, what their deductible is, why a claim was denied, expected out-of-pocket cost, provider network status and what an EOB means. Support agents needed to search across benefit documents, policy information, claims data and knowledge bases to answer them.

Architecture: Member/CSR → AI Assistant → Intent Detection → RAG → Member/Claims APIs + Vector DB → LLM → Guardrails → Response

Technologies: Azure OpenAI / OpenAI, LangChain, Python, FastAPI, Azure AI Search, Vector Database, API Gateway, Azure Functions / AKS, Application Insights, LangSmith

- Built MemberAssist AI, a RAG-based GenAI assistant that retrieves a member's relevant plan, benefit and claims information and explains it in plain language for support agents and members.
- Implemented intent detection to route member questions — coverage, deductible, claim denial reason, cost-share, network status, EOB explanation — to the appropriate retrieval and API path.
- Developed RAG retrieval over benefit documents, policy information and internal knowledge bases, combined with real-time lookups against member and claims APIs.
- Example: for a denied-claim query, the assistant retrieves the claim, denial reason, member benefits and applicable policy, then generates a plain-language explanation (e.g., missing prior authorization) along with next steps such as contacting the provider or reviewing appeal options.
- Generated embeddings and indexed benefit guides and policy documents in a vector database for semantic search.
- Built FastAPI endpoints and API gateway integration to expose the assistant to CSR-facing tools.
- Applied guardrails and grounding checks so responses stayed accurate to member-specific plan data, avoiding incorrect coverage or claim-outcome statements.
- Used LangSmith for tracing and debugging LLM calls, and Application Insights for production monitoring.

JAVA DEVELOPMENT — Claims & Member Services Backend (Java)

Context: Began as a Java backend developer on Anthem's claims and member services platform, building and maintaining the core services that GenAI applications would later integrate with, before transitioning into the GenAI engineering team.

Technologies: Java, Spring Boot, REST APIs, SQL, JUnit, Mockito, Git, Jenkins, Agile

- Developed and maintained backend microservices using Java, Spring Boot and REST APIs for claims processing and member service workflows.
- Designed and optimized SQL queries and stored procedures for claims and member data, and wrote unit/integration tests using JUnit and Mockito to ensure service reliability.
- Collaborated with QA and business teams to troubleshoot production issues, supporting release cycles via Git and Jenkins CI/CD pipelines in an Agile environment.
- Transitioned into the GenAI engineering team, applying backend and API expertise to LLM-powered application development.

EDUCATION

Bachelor of Technology (B.Tech)
JNTUK University • 2023

CERTIFICATIONS

- Databricks Certified – Generative AI
- LangChain Certification
- LangGraph Certification